

A Finite-Panel Audit of Context Selection in a Frozen VMem Configuration

HKUST CSIT6910 independent research project

18 September 2026

Question. A released view-generation system selects four historical frames to condition each generation. On a fixed panel it delivers only three *distinct* frames into those four slots in 10 of 14 windows, because two independently seeded selections return the same frame. Does that wasted slot cost image quality?

Intervention. Replace the redundant slot with the pipeline’s own next distinct ranked candidate, in place, changing nothing else. Retention threshold +0.20 dB, fixed in writing before any repair output was scored.

Result. -0.016 dB, across the 8 of 10 duplication-affected windows in which the prespecified policy offers a replacement candidate. The no-op control passed byte-identically.

Decision. The performance explanation is discarded and this branch is closed. No further generation is planned.

1 Outcome and stopping decision

A prospectively specified intervention was run and did not meet its criterion.

The released selector delivers only three distinct frames into four context slots in 10 of 14 evaluated windows. The proposed explanation was that this wasted slot materially costs image quality. The test was an in-place replacement of the redundant slot with the pipeline’s own next distinct ranked candidate, with a retention threshold of +0.20 dB fixed in writing before any repair output was scored.

Measured: -0.016 dB in mean across the 8 of 10 duplication-affected windows in which the prespecified policy offers a replacement candidate (SD across windows 0.309, 3 of 8 positive). In the remaining 2 affected windows the intervention could not be instantiated under the prespecified candidate-selection policy, because that policy provided no fourth distinct candidate: requesting up to seven entries still returned four, holding three distinct frames (record in Appendix B). That is a property of the prespecified read policy, not a demonstration that those windows are untestable under any possible design. The no-op control passed: all 8 window–seed executions in the 4 unaffected windows reproduced the sealed output byte-identically.

The duplication-as-performance-explanation is **discarded**. The threshold was not lowered, the replacement policy was not changed, and no favourable subgroup was elevated. **This branch is closed.**

Failure to meet a +0.20 dB criterion is *not* proof of no effect. The source-level cause of the duplicate selection remains established; only its proposed material PSNR consequence does not. Two results were also withdrawn during the work and are recorded in Section 8.

2 Scope and the quantity measured

Data. Two exposed development sequences, RGB-D Scenes v2 scene_13 and scene_14, seq-01. These sequences are exposed: they shaped the hypotheses during this project. Their target frames are excluded

from conditioning, but the sequences are *not* independent held-out evaluation data and the term “held-out” is not used for them anywhere.

Panel. 14 fixed windows (7 window starts $\times$ 2 sequences; window start 0 fails retrieval in both sequences and is excluded with that reason). Bank = start + {0, 5, ..., 55}, 12 frames. Targets = start + {60, 75, 90, 105}, i.e. forward extrapolation beyond the bank. Two fixed seeds (42, 7). One dependency group. One frozen consumer.

Unequal horizons. The fixed-offset arm may use frames only up to offset +45; the retrieval bank extends to +55. Retrieval therefore has a candidate-availability advantage. This is *not* a guarantee of any performance ordering: a frozen generator’s output is not monotone in candidate proximity, and this selector carries no dominance guarantee.

Metric. RGB PSNR only. Reference frames are loaded at native 480 $\times$ 640, converted to float, area-resized to 576 $\times$ 768, cropped to columns 96:672, and quantised to uint8. Predictions are mapped from $[-1, 1]$ to $[0, 1]$ when needed and quantised identically. **MSE is accumulated over all four targets of a window–seed before a single PSNR is taken.** No geometry metric, no perceptual metric, no human study.

Aggregation. Seeds are controlled stochastic replications, not independent scenes. The two seeds are averaged *within a window first*; all reported statistics are then window-level.

Inference. Windows are a finite panel, not a sample from a population. The quantity measured is the mean over exactly these windows and seeds. **No standard error, t statistic, confidence interval, significance test, equivalence test or bootstrap appears in this report.** SD is reported solely as a description of heterogeneity across the panel.

What is compared. Every arm receives real historical frames. The principal baseline comparison is between retrieval-selected historical context and fixed-offset historical context. The remaining contrasts compare state initialisation or context-selection variants against one another. **None of them is a comparison of memory versus no memory.**

3 Software and execution provenance

Consumer. VMem ([arXiv:2506.18903](https://arxiv.org/abs/2506.18903)), public repository runjiali-rl/vmem. Frozen: no training, no fine-tuning, no new weights, no modification of the upstream source.

Pinned source verified against the live public repository on 18 September 2026. Public HEAD commit 39291e4f272f6b4f270691d930926ab5930f942e (25 July 2025). The project’s pinned navigation.py is byte-identical to the file at public HEAD. Full hashes in Appendix C. No local patch was applied to upstream code at any point.

Complete generation call graph of the release. navigation.py is the only caller of generate_trajectory_frames, and it has *three* call sites:

line	method	reached from	effective NMS setting
185	move_backward	app.py:219	explicit False $\rightarrow$ disabled
234	move_forward	app.py:215	explicit False $\rightarrow$ disabled
321	_turn	app.py:208,210 via turn_left/right	no argument $\rightarrow$ config true $\rightarrow$ enabled

get_context_info has exactly one internal caller, pipeline.py:1249. **The released demo therefore uses both retrieval settings on the same pipeline object, selected by which action the user takes.**

Retrieval state semantics. In get_context_info: the NMS-disabled branch executes self.initial_threshold = 1e8 unconditionally and then reads it; the NMS-enabled branch assigns the attribute only when the

pipeline holds exactly five frames; at any other bank size the enabled branch reads whatever value is present. `reset()` does not clear the attribute.

Consequence, stated as static reachability only. A native interaction of the form `initialize` $\rightarrow$ `move_forward` $\rightarrow$ `move_forward` $\rightarrow$ `turn_left` leaves the bank at a size other than five when the turn executes, so the enabled branch consumes the `1e8` written by the preceding movement. The threshold a turn uses depends on whether the user moved first. **The demo was not run; this is a source-level reachability statement plus the selector’s own semantics, not a measured demo behaviour.**

Evaluation entry point. At public HEAD the repository’s non-extern/ Python files are exactly `app.py`, `navigation.py`, `modeling/*` and `utils/*`. **No evaluation or benchmark script is present.** The configuration that produced the paper’s tables therefore cannot be determined from the release, and this report makes no claim about it.

Job ledger. 595599 order-invariance gate (zero diffusion); 595614 leak-regime census (zero diffusion); 595625 disabled-branch threshold-independence (zero diffusion); 594957 sealed three-arm run; 595887 clean NMS-on arm; 595891 refill repair arm; 595902 in-place repair arm.

4 Arms and prospective specifications

arm	context	corresponding released behaviour
<code>static</code>	fixed offsets <code>+0, +15, +30, +45</code> ; no retrieval call	none — a project-defined baseline
<code>memory_nms_off</code>	pipeline retrieval, NMS disabled	the movement path
<code>memory_nms_on</code>	NMS enabled under an inherited <code>initial_threshold = 1e8</code>	turning <i>after</i> a movement
<code>memory_nms_on_clean</code>	NMS enabled from an isolated state	turning, with an isolated threshold assigned at the five-frame priming step. “Clean” is this operational definition only; it does not assert the configuration the authors intended or used.
<code>refill</code>	$(a, a, b, c) \rightarrow (a, b, c, d)$	none — repair variant, source-faithful fill order
<code>inplace</code>	$(a, a, b, c) \rightarrow (a, d, b, c)$	none — repair variant, primary

The correspondence is *structural, not exact*: in the demo the bank grows as generated frames are written back, whereas this diagnostic holds a fixed 12-frame bank.

Repair specification and when it was fixed. `REPAIR_SPEC_PREDECLARED_20260918.md` was written before any repair output was scored, after an external review identified that the `refill` mapping mixes a content change with a position change. It names `inplace` primary, retains `refill` as the source-faithful variant, fixes the `+0.20 dB` threshold, fixes the eligibility rule, and records the `8/4/2` window accounting. The already-sealed `refill` arm was *not* deleted or rewritten.

Candidate-list provenance, verified rather than argued. The ranked candidate list is read out of the pipeline, not re-implemented. Because `pipeline.py:492` computes `num_retrieved_frames = min(context_num_frames + 10, len(timestep_weights))`, requesting a longer selection could in principle enlarge the candidate pool, so a matching four-element prefix would not certify the remainder. The list was therefore obtained at `context_num_frames` 5, 6 and 7 — three different pool requests — with the query centre held fixed by passing `target_c2ws[-1:]`. **In every evaluated window the lists were exactly nested**, e.g. `[11, 11, 10, 9, 8]  $\subset$  [11, 11, 10, 9, 8, 3]  $\subset$  [11, 11, 10, 9, 8, 3, 4]`, with the shipped `[11, 11, 10, 9]` as prefix, and the distinct-frame count never exceeded the bank size. This is a statement about exactly those requests and windows, not a theorem about all candidate pools.

Leak-regime census, fixed before any scoring. Every window was classified by whether the leaked threshold changed the padded context ids: `NULL` 2, `PERMUTATION` 4, `CONTENT` 8. Slot 0 was

identical in 14 of 14 windows, as the source requires — `sorted_frames[0]` is appended before the threshold loop and never reads the threshold, so the leak cannot move the ray-reference camera.

Repair window manifest. Full 14-window accounting in `REPAIR_WINDOW_MANIFEST_20260918.md`: 4 no-op controls + 10 duplication-affected, of which 8 repaired and scored and 2 excluded for having no replacement candidate. The 8 repaired windows are *not* the same set as the 8 CONTENT windows of the leak census; they differ in four entries.

5 Results and validity checks

All contrasts signed, window-level, seeds folded first.

5.1 Context configuration contrasts ($n = 14$ windows)

contrast	mean	SD	windows positive
<code>memory_nms_off – static</code>	+0.242 dB	1.270	8/14
<code>memory_nms_on_clean – static</code>	–0.485 dB	1.489	6/14
<code>memory_nms_on_clean – memory_nms_off</code>	–0.726 dB	1.210	5/14
<code>memory_nms_on_clean – memory_nms_on</code>	+0.245 dB	0.711	6/14

Median of the first row is +0.385 dB; range –2.073 to +2.055.

Absolute arm means, **restricted to the same 14-window paired set** (14 windows $\times$ 2 seeds = 28 executions per arm), so that differences of the absolute means reproduce the paired contrasts exactly:

arm	executions	mean PSNR	SD across windows
<code>memory_nms_off</code>	28	14.637 dB	1.718
<code>static</code>	28	14.395 dB	1.558
<code>memory_nms_on_clean</code>	28	13.911 dB	1.838
<code>memory_nms_on (leaked)</code>	28	13.666 dB	1.789

Check: $14.637 - 14.395 = +0.242$; $13.911 - 14.395 = -0.485$; $13.911 - 14.637 = -0.726$; $13.911 - 13.666 = +0.245$. Each matches the corresponding paired contrast above.

The static arm was also generated for two further windows (window start 0 in each sequence) where retrieval terminates with a failure, so no memory arm exists to pair with. Its mean over all 32 executions is 14.560 dB. **That figure appears here only to document the wider generation set; it must not be differenced against any arm measured on the 14-window panel**, because the window sets differ. All contrasts in this report use the 14-window paired set.

5.2 Leak effect, stratified by the pre-fixed census

stratum	n	mean of clean – leaked	SD
NULL (identical selection)	2	+0.000 dB	0.000
PERMUTATION (same multiset, different order)	4	–0.015 dB	0.015
CONTENT (different multiset)	8	+0.436 dB	0.917

The NULL stratum is exactly zero because the two arms received byte-identical consumer input and produced byte-identical output — verified by sha256 on all four runs. That is the validity gate for reusing sealed outputs instead of regenerating them, and it passed.

The PERMUTATION stratum is the informative one. Holding the context multiset, cameras, intrinsics, normalisation input and slot-0 ray gauge all fixed and changing only slot order moved PSNR by -0.015 dB. In this regime, the leak’s entire effect is carried by *which frames* were selected, not by the order they occupy. Scope limit: this is order variation among orderings the retrieval itself produced, on four windows; a non-zero difference is expected from reordered floating-point reductions alone, and the finding is that the magnitude is negligible here, not that the computation is invariant.

5.3 Repair arms ($n = 8$ of 10 duplication-affected windows)

contrast	mean	SD	windows positive
inplace – memory_nms_off	-0.016 dB	0.309	3/8
refill – memory_nms_off	-0.032 dB	0.304	3/8
inplace – refill (pure order)	$+0.016$ dB	0.037	5/8

These three are algebraically linked: $(-0.016) - (-0.032) = +0.016$ up to rounding. The order comparison is *not* independent confirmation of the other two.

5.4 Validity checks, with their exact coverage

- **Order-invariance gate** (job 595599, zero diffusion). The name refers to invariance of the *retrieval selection* to the order in which arms are executed. It says nothing about whether generated output is invariant to context slot order, which is a different question addressed in Section 5.2. The generation worker’s own isolation routine makes the effective threshold, the ordered frame ids and the padding multiplicities equal between “clean arm alone” and “other arm first, then isolate, then clean arm”. 11/11 checks, including a non-vacuity assertion that the test can detect the contaminated ordering it guards against. Covers 3 windows of scene_13.
- **Disabled-branch threshold independence** (job 595625, zero diffusion): forcing `initial_threshold` to the primed percentile, $1e8$, $1e-9$, 0.0 , and deleting it, leaves the NMS-disabled selection unchanged. 6/6 windows. Deleting does not raise, because the disabled branch assigns before reading — which is also why it is the leak source.
- **NULL byte-identity gate**: 4 window–seed executions (2 NULL windows $\times$ 2 seeds). All sha256-identical between the clean and leaked arms.
- **No-op byte-identity gate**: 8 window–seed executions (4 unaffected windows $\times$ 2 seeds). All sha256-identical to the sealed shipped output.
- **Seal verification**: all 28 new outputs and all 88 sealed outputs re-hashed and matched before any target RGB was opened.
- **Historical linkage**: the contaminated ordering was reproduced 28/28 byte-identically against the sealed run, and the census independently reproduced the sealed leaked selections frame-for-frame.

Each gate certifies exactly the cases it covers and nothing beyond them.

6 Interpreting the failed repair

Three distinct claims must be kept apart.

The cause of three-distinct-frame delivery is established. With NMS disabled the selection is seeded with `sorted_frames[0]` (the surfel-nearest candidate) and then `len(self.c2ws) - 1` (the most recent stored frame). **In 10 of the 14 evaluated forward-extrapolation windows those two coincide.** When it occurs, the duplicated index counts toward the requested context length, so the fill step cannot recover the slot and only three distinct frames reach four slots. Forward extrapolation is the setting in which this was observed; it is *not* shown to be a sufficient condition, since four of the fourteen forward-extrapolation

windows do not exhibit it. It is visible directly in the ranked indices: [11, 11, 10, 9], index 11 being simultaneously the nearest candidate and the most recent frame.

Whether repairing it improves PSNR was tested and the answer did not meet the criterion. -0.016 dB over 8 of 10 affected windows.

Whether repeated information actively harms generation was not tested and cannot be answered here. That would require an absent-slot control $(a, -, b, c)$ with a genuinely absent context slot and everything else preserved. Under a four-frame selection-only intervention that leaves the conditioning interface untouched, no such control exists: a blank image is still an input, any other existing image changes multiplicity, and removing a slot changes sequence length and target placement. No substitute was attempted and none is claimed.

Why a near-zero mean is not a near-zero effect. With mean -0.016 dB and SD 0.309 over 8 windows, per-window effects of a few tenths of a dB in both directions cancel. Similar PSNR values also do not imply similar images: two outputs can have comparable error against the target while differing from each other. Only the byte-identity gates establish exact sameness, and only for the cases they cover.

Why the interventions do not identify a distance-sensitivity mechanism. All three figures below are signed differences, not magnitudes. Reordering a multiset: -0.015 dB (4 windows, clean minus leaked). Replacing a duplicate with the next ranked nearby frame: -0.016 dB (8 windows, in-place minus movement-path). The NMS policy contrast: -0.726 dB (14 windows, clean NMS-on minus movement-path). Those three have different interventions, different comparison arms and different window sets, and “same multiset”, “nearby frame” and “distant frame” are not three levels of a controlled variable — a larger frame index separation does not establish a proportionally larger change in pose, visible content, or the conditioning tensors. A competing explanation, that this is simply a monotone response to larger input perturbation, is not excluded. **The defensible statement is only that the tested NMS policy produced a larger negative mean contrast than the other tested interventions, each on its own window set.**

7 Relation to prior work, and limitations

Read at primary source on 18 September 2026. Each of the following occupies a claim this project might otherwise have made, and each is therefore recorded as a reason *not* to claim novelty.

- **VRAG** ([arXiv:2505.21996v4](#)): “The History Buffer method performs poorly, with an SSIM score of 0.188, indicating that naive historical frame retrieval without effective in-context training fails to maintain long-term consistency.” Occupies “historical retrieval need not beat recent-window conditioning”. Its own 300-frame table is not uniformly negative across metrics, so it must not be cited as a blanket negative.
- **Context as Memory** ([arXiv:2506.03141](#)): context-selection ablation, Random 17.70/17.07, FOV+Random 19.17/17.47, FOV+Non-adj 20.11/18.19. A published redundancy-selection ablation in the same units used here.
- **RAGME** ([arXiv:2504.06672](#)): named deduplication of near-duplicate retrieved videos in retrieval-augmented video generation.
- **LongLive-RAG** ([arXiv:2606.02553](#)): retrieval collapsing toward temporally local neighbours, with a loss that suppresses redundant local similarity.
- **COVRAG** ([arXiv:2606.02479](#)): finer geometric evidence with independent selection (MEt3R 0.149, LPIPS 0.210) is worse than coarser evidence with independent selection (0.141, 0.198); residual selection recovers it (0.100, 0.156).

Two further sources, **PRoPE** ([arXiv:2507.10496](#)) on camera reference-frame dependence and **TTAB** ([arXiv:2306.03536](#), ICML 2023) on episodic versus accumulated evaluation state, closed two directions that were considered and abandoned before the repair experiment. They are recorded in Appendix D.

Relation to VMem’s own evaluation. The paper describes non-maximum suppression as part of its method: “To avoid oversampling repeatedly visited regions, we apply a non-maximum suppression algorithm that reduces redundancy in memory and promotes broader scene coverage among the top- K .” Its memory demonstration is §4.3, “Long-term view generation with revisitations”; its $K = 4$ selector ablation reports temporal selection at 7.52 dB against VMem at 14.82 dB. **This report’s panel contains no revisitation: every target lies beyond the end of the conditioning bank. This work therefore does not reproduce, test, or contradict that experiment, and no absolute score from this panel is comparable with one from it.**

Limitations. One consumer cannot establish prevalence across a model family. One dependency group cannot establish an average effect across environments. Exposed sequences cannot later serve as untouched validation. Additional seeds, windows or replays would supply no independent scenes and would not widen the metric’s scope. PSNR contrasts establish RGB prediction-error differences under the stated conditions and nothing about perceived realism, long-horizon scene consistency, or world model usefulness.

8 Corrections, disposition, and artifact index

8.1 Corrections made during this work, all retained in the record

1. **A contaminated arm was interpreted as a memory result.** The sealed NMS-enabled arm ran under a threshold inherited from a preceding NMS-disabled call. Its -0.729 dB figure was withdrawn as an estimate of the independently initialised effect and replaced by `memory_nms_on_clean` at -0.485 dB. The original number is retained as the result of a specified order-dependent execution.
2. **A positive mean was described with the wrong sign.** The $+0.242$ dB contrast was written as “does not beat fixed-offset context” and “a bounded negative result”. Both retracted.
3. **Inferential statistics were reported for a finite panel.** SE 0.340 and t 0.71 withdrawn; SD retained as heterogeneity only.
4. **“Held-out” was used for exposed sequences.** Narrowed throughout.
5. **The release’s call graph was traced by parameter name rather than by function name.** This produced four false claims — that every released call disables NMS, that the config default is unreachable, that NMS-on is not shipped behaviour, and that the leak belongs only to comparative harnesses. All four are retracted; Section 3 gives the complete three-site call graph. The procedural lesson is recorded in the project principles: a parameter-name search finds only the sites that override it, and the count of overrides is never the count of call sites.
6. **A repair mapping mixed a content change with a position change.** The refill arm was retained and reported, and the in-place arm was added and made primary, with the specification fixed before scoring.
7. **A repair denominator was stated without its exclusions.** Corrected to “8 of 10 duplication-affected windows”, with the 2 exclusions and their structural reason documented.

8.2 Disposition

The research branch is closed. No further generation is planned. No novelty claim is supportable from this work and none is made. `new_method_validated` remains `false`; `novelty_authorization` remains `NONE`.

An open question to the upstream authors is prepared but **not sent**: whether the per-action retrieval setting in `navigation.py` is intentional, and which entry point and configuration produced the selector ablation. It is phrased as a question, scoped to the exact commit and files inspected, carries no inference about intent, and does not attach this project’s measurements.

9 Addendum, 18 September 2026: post-hoc target-wise scoring note

Added after the report was finalized. It does not alter the primary results, the prespecified scoring rule, or the decision to close the duplicate-slot repair branch.

Post-hoc target-wise analysis. At target offsets +60, +75, +90, and +105, the mean framewise PSNR contrasts for NMS-off minus fixed-offset context were +2.950, +0.398, −0.180, and −0.321 dB, respectively, with seeds averaged within windows. Their arithmetic mean is +0.712 dB. This uses a different aggregation from the prespecified four-frame pooled-MSE PSNR contrast of +0.242 dB and is not its additive decomposition. The profile is descriptive and does not isolate temporal distance, target-slot position, or context clustering as a cause.

The two aggregations differ because the prespecified score pools MSE across the four targets before taking one PSNR, whereas the framewise figures take one PSNR per target and average. The resulting gap is non-negative and **arm-dependent** — measured here as +0.227 dB for the fixed-offset arm and +0.697 dB for the NMS-off arm — so contrasts computed under the two aggregations are not interchangeable. **Section 5 remains the primary result and its scoring rule is unchanged.**

A The 14-window table

Window means in dB (the two seeds of a window averaged first). *distinct* is the number of distinct frames the shipped movement-path retrieval delivered into the four context slots.

scene	win	static	nms_off	nms_on (leaked)	nms_on (clean)	inplace	refill	dist.	repair status
scene_13	w050	13.027	15.082	12.485	15.066	15.075	15.066	3	repaired
scene_13	w100	15.077	16.115	12.374	12.131	—	16.115	4	no-op control
scene_13	w150	11.390	11.895	11.106	11.107	—	11.895	4	no-op control
scene_13	w200	15.379	13.807	13.789	13.789	—	—	3	not instantiable
scene_13	w250	14.791	15.338	14.064	14.822	14.844	14.820	3	repaired
scene_13	w300	17.155	17.025	15.462	15.433	17.208	17.210	3	repaired
scene_13	w350	15.419	15.071	13.136	13.318	15.307	15.320	3	repaired
scene_14	w050	14.524	14.788	14.233	14.420	14.447	14.404	3	repaired
scene_14	w100	14.969	16.622	16.641	16.641	—	—	3	not instantiable
scene_14	w150	11.695	13.669	11.770	11.765	—	13.669	4	no-op control
scene_14	w200	15.292	16.270	16.256	16.417	—	16.270	4	no-op control
scene_14	w250	15.236	14.832	15.434	15.335	14.687	14.722	3	repaired
scene_14	w300	14.289	13.187	13.259	13.230	13.185	13.170	3	repaired
scene_14	w350	13.290	11.217	11.315	11.274	11.663	11.578	3	repaired
mean		14.395	14.637	13.666	13.911				

The refill column is present in the four no-op windows because that arm generated them as its byte-identity control; inplace skipped them by design, and the sealed shipped output is the control for both. Where a leaked and a clean value are identical to three decimals (w200 scene_13, w100 scene_14) the two arms received byte-identical input: these are the NULL-stratum windows, and their outputs are sha256-identical, not merely close.

B The two windows where the repair could not be instantiated

The candidate list is read from the pipeline by requesting a longer selection. In these two windows the request returns four entries no matter how many are asked for, and those four hold three distinct frames, so the prespecified policy supplies no fourth distinct candidate.

window	requested	returned indices	frames (distinct)
scene_13 w200	5, 6, 7	[11, 11, 10, 9]	[255, 255, 250, 245] (3)
scene_14 w100	5, 6, 7	[11, 11, 10, 9]	[155, 155, 150, 145] (3)

scene_13 w200: bank [200, 205, . . . , 255], shipped context [255, 255, 250, 245], duplicate at slot 1, replacement candidates []. scene_14 w100: bank [100, 105, . . . , 155], shipped context [155, 155, 150, 145], duplicate at slot 1, replacement candidates [].

This records how the “no candidate” determination arose. It establishes that the intervention as specified has nothing to insert; it does *not* establish that no fourth frame could ever be useful in these windows under a different design. For contrast, a repairable window (scene_13 w050) returns a list that does extend: $[11, 11, 10, 9, 8] \subset [11, 11, 10, 9, 8, 3] \subset [11, 11, 10, 9, 8, 3, 4]$ at requests 5, 6, 7.

C Run configuration

Source and checkpoint identity

Upstream repository runjiali-r1/vmem, public HEAD 39291e4f272f6b4f270691d930926ab5930f942e (25 July 2025). **No local patch was applied to upstream code at any point.** navigation.py is byte-identical to the file at public HEAD.

file	sha256
navigation.py	6d267365d5dccc9f7cd6d535f19f80521f60a9634dfb35b9af398407389fcf
modeling/pipeline.py	90a45f452a4f734b3371fb526c566162026e5d900e0c8204709dd57f44ef1d7e
app.py	02144b511a89c5619a380b0d08e2f62ce0946d7185046d4fd035d9caad28cc31
configs/.../inference.yaml	8d849588016935573a22ef6aaee567f71125ca4d3bdf18f51e3552a64be9fea3
vmem_weights.pth	675dc486a02ea06ecf8b6ab0cf4ef88c92298751b2daac9f965c59871fcb7fe4
cut3r_512_dpt_4_64.pth	45f7e98a0a64dbeb54901ae2b878cd8cd125f20a4497316483f0bd6f109f8103
open_clip_model.safetensors	0084e75319a50ad85ef45377bad5bc38f2f58824459eb690048d51c9f8863be5
diffusion_pytorch_model.safetensors	a1d993488569e928462932c8c38a0760b874d166399b14414135bd9c42df5815
VAE config.json	92d3dfb746fca211a2c9e019e285f8597412211728dce3c5bcf4eda0f2d62e7e
container image digest	63a18dd805367dacfb077aeced8384ab2fb569598ec5f5f5220c3f90a5c23650
container SIF	5a79221373914393c844cc92c32c89e722003591431f3545fc674c0739c59dd0

Execution settings

item	value
apptainer flags	-nv -containall -no-home -cleanenv; read-only binds; offline HF/transformers
hardware	HKUST SuperPOD, 1 × H800 per job
sampler	create_samplers(guider_types=1, ...) over DDPMDiscretization, DiscreteDenoiser(num_idx=1000)
diffusion steps	50 (inference_num_steps)
guidance	cfg = 2.0, cfg_min = 1.2
frames per forward	8 = 4 context + 4 target; input_masks = [T, T, T, T, F, F, F, F]
resolution	576 × 576, latent C = 4
dtype	fp32 parameters with CUDA autocast (upstream default)
context_num_frames	4 (native); 5, 6 and 7 used <i>only</i> for the zero-diffusion candidate-list read
use_non_maximum_ suppression	config true; effective value per arm as recorded in Section 4
seeds	42 and 7, set on random, numpy, torch, torch.cuda before each generation
threads	torch.set_num_threads(8), set_num_interop_threads(1)

D Directions closed before the repair experiment

Recorded because they explain why other lines were not pursued, not because they bear on the result.

- **Camera reference-frame dependence.** PRoPE ([arXiv:2507.10496](#)): “sensitive to the arbitrary choice of reference frame, which can hinder generalization.” Independently, the selector’s own source shows `sorted_frames[0]` is appended before the threshold loop and never reads the threshold, and slot 0 was identical in 14 of 14 windows, so the state leak provably cannot move the ray-reference camera. The direction was dead on both counts.
- **Evaluation state and resetting.** TTAB ([arXiv:2306.03536](#), ICML 2023) distinguishes episodic adaptation from accumulated state. State isolation therefore remains a necessary control in this work, not a contribution.
- **Source-conflict and future-geometry selection candidates** (project-internal, SOCF-A, FGB-SI) were stopped before execution: they proposed selection policies before anything was established about what the consumer responds to.

E Project governance flags

`new_method_validated = false`. `novelty_authorization = NONE`. No statement in this report claims human approval for a method, a novelty finding, or a release. The sequences used are exposed development data; their target frames are excluded from conditioning but the sequences are not independent held-out evaluation data.

F Delivery bundle

Accompanying bundle/, so that any number in this report can be traced without entering the working tree.

file	content
S113_SCORES.json	per window–seed scores, all four context arms, seal verification, NULL byte-identity record, leak strata
SLOT_REPAIR_SCORES.json	per window–seed scores for both repair arms, no-op byte-identity record, verdict
LEAK_REGIME_CENSUS.json	pre-scoring window classification and per-window selections
SLOT_CONTROL_RECEIPT_inplace.json	in-place arm receipt: candidate provenance checks, repair mapping, output sha256
SLOT_CONTROL_RECEIPT_refill.json	refill arm receipt, same fields
NMS_ON_CLEAN_RECEIPT.json	clean-arm receipt, isolation assertions, threshold recorded at selection
NMS_RECEIPT.json	sealed three-arm run receipt (job 594957)
REPAIR_SPEC_PREDECLARED_20260918.md	the repair specification, fixed before any repair output was scored
REPAIR_WINDOW_MANIFEST_20260918.md	the 14-window accounting
WEIGHTS_SHA256.txt	checkpoint hashes as listed in Appendix C
MANIFEST.sha256	sha256 of every file above; verify with <code>shasum -a 256 -c MANIFEST.sha256</code>

The JSON receipts and score files were produced on the cluster and copied without modification; each was confirmed byte-identical to its cluster original on transfer.